

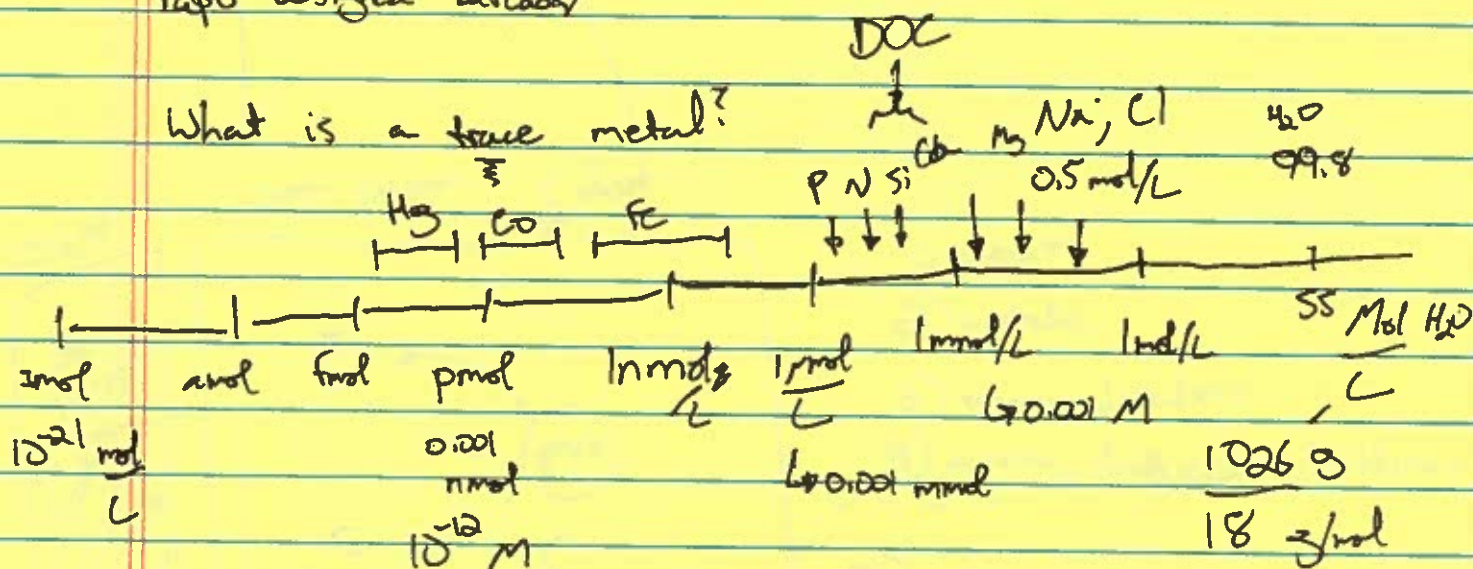
Chemical Oceanography 3/2/21

Trace metal cycles in the oceans

- 1) Profiles & processes
- 2) microscale chemistry
- 3) metal limitation

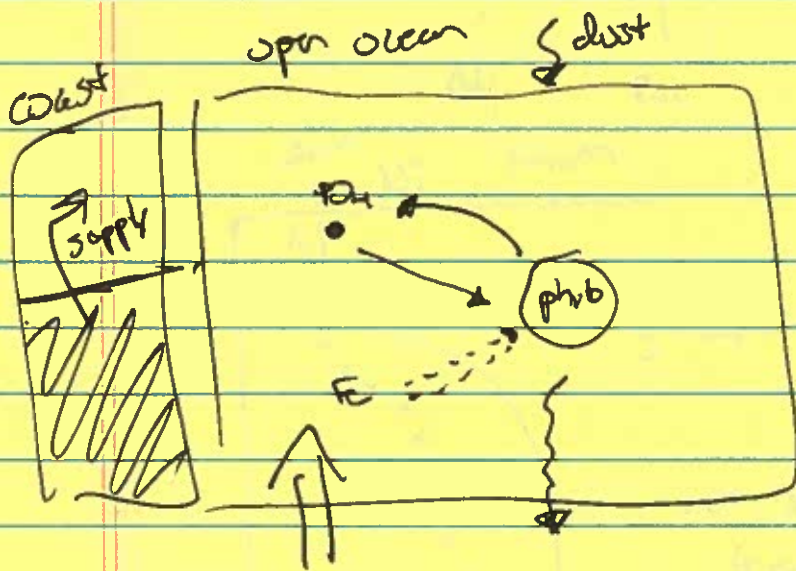
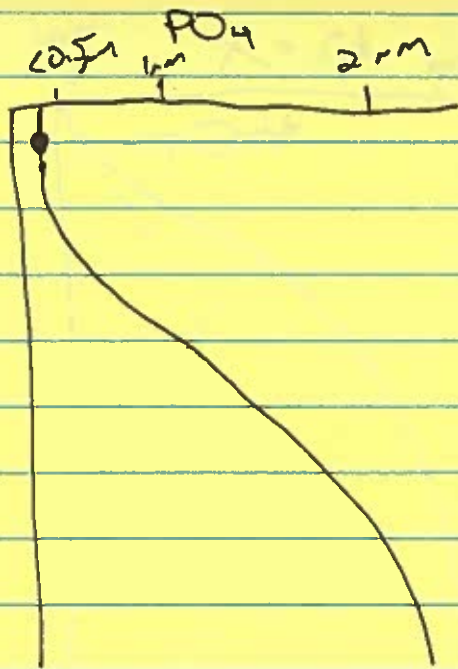
Problem set assigned Thursdays
 Paper assigned already

What is a trace metal?

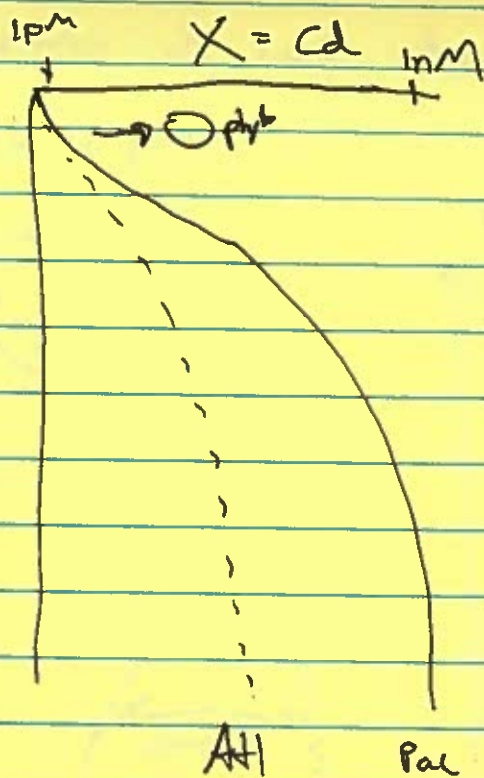


What is trace?

$X \ll \text{major ions}$
 $X \ll \text{DOC}$



- 1) export
- 2) uptake
- 3) remineralization
- 4) mixing / advection / diffusion
- 5) adsorption
- 6) supply from sediments, dust
glaciers, sea ice, rivers
hydrothermal vents



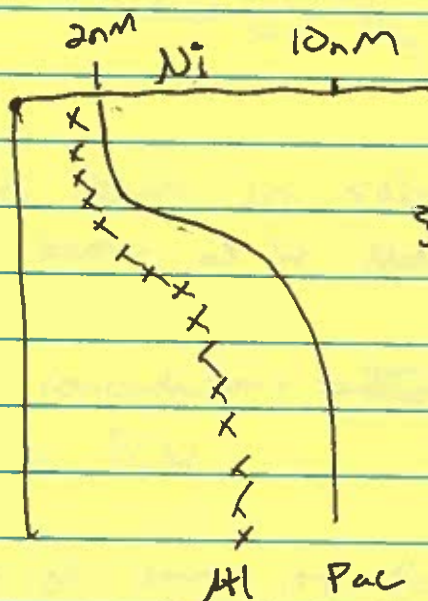
- 1) uptake
- 2) adsorption (onto phytoplankton)
- 3) export
- 4) regeneration

Cd has a perfect nutrient like profile

but Cd has not bio rate

↳ used to reconstruct PO_4

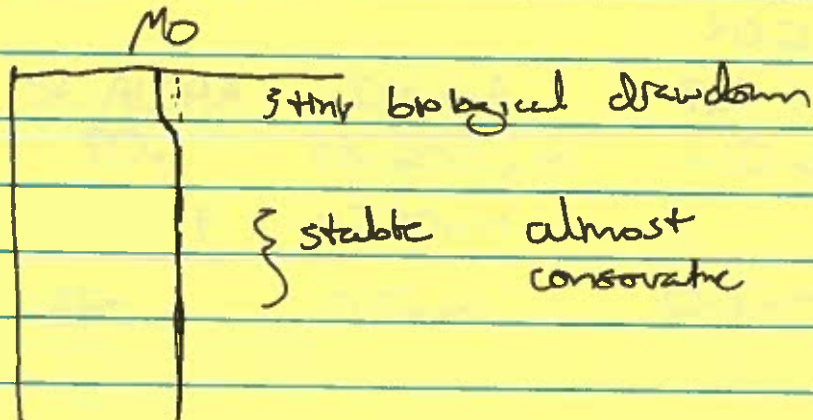
bc Cd^{2+} can substitute for Ca^{2+} in $CaCO_3$



↳ sharper (Wick's drawing mistake)

not drawn down totally

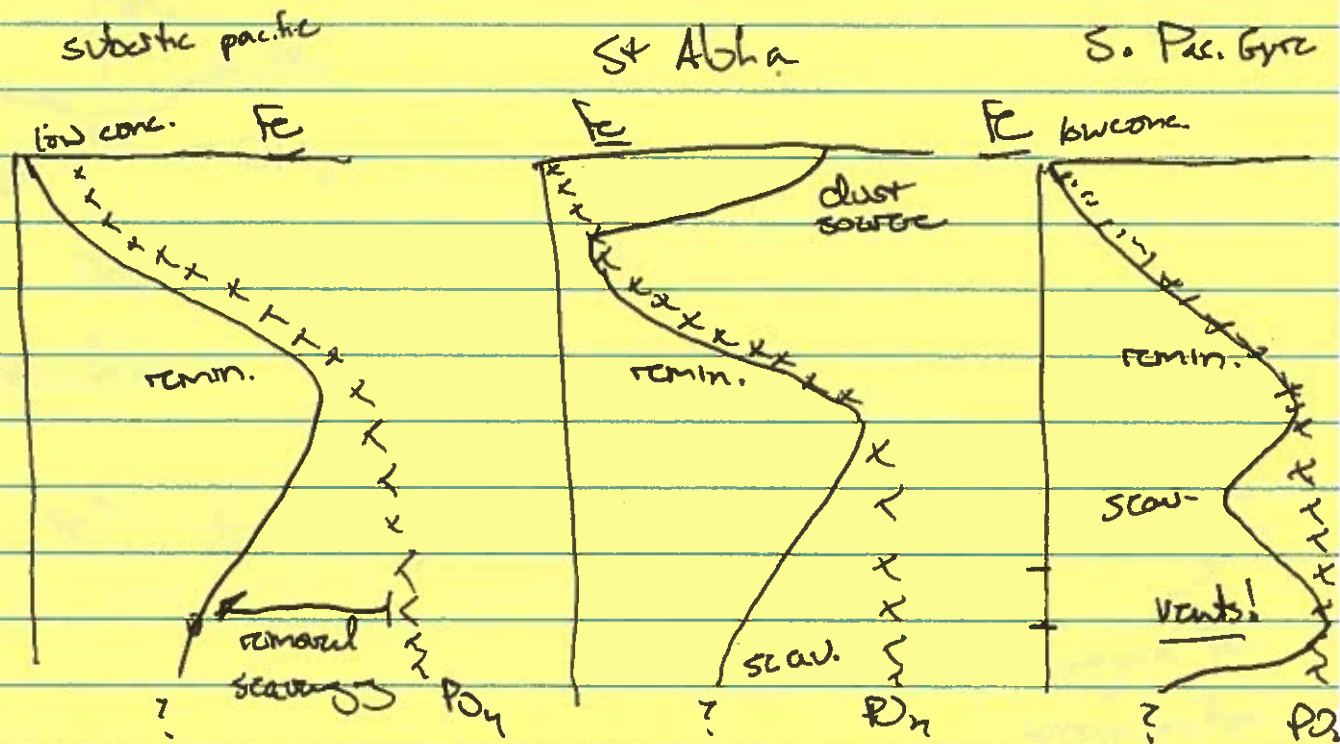
↳ explanation: organisms don't use very much Ni



3 m biological drawdown

{ stable almost conservative

3 profiles of same element at diff. parts of the Pacific



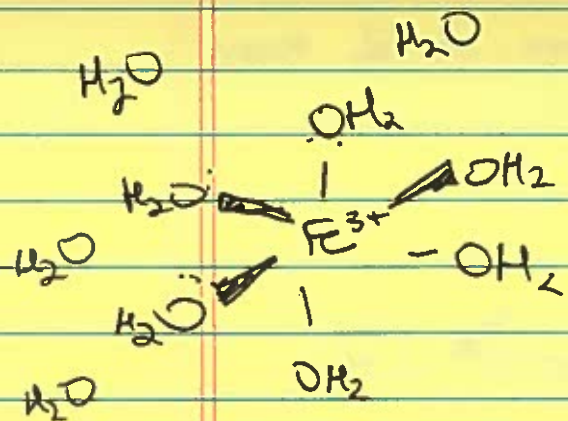
- drawn down in sub arctic, S.P.

- bigger source at St Aloha

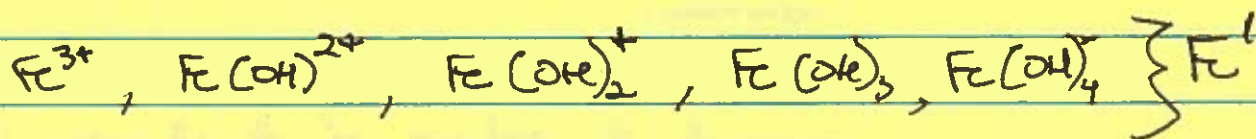
- similar concentrations to ~~PO₄~~ in all places
less than PO_4

a lot of iron in the ocean is "particulate"

			> 0.2 μm	typical for low residence time
	St. ALOHA	Dissolved	Part	
phosphorus	PO_4	50-200 nM	~10 nM	} $dPO_4 \gg pPO_4$
	org P	~200 nM		
iron	dFe	0.5 nM	0.5 nM	} $3 pFe \sim p dFe$
				faster exchange faster residence time



octahedron

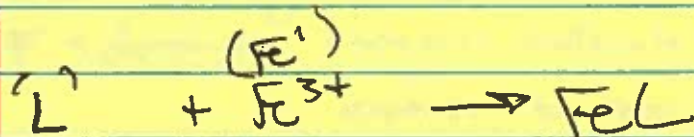


inorganic Fe
concentration

relatively fast exchange

$$[X] \ll \text{DOC}$$

$$\text{Fe} \ll \text{DOC}$$



ligands

an organic molecule

of high affinity for iron

$$K_{\text{Fe}^I \rightarrow \text{FeL}} = \frac{\text{FeL}}{\text{Fe}^I \cdot L} \rightarrow \frac{\text{FeL}}{\text{Fe}^I} = K \cdot L$$

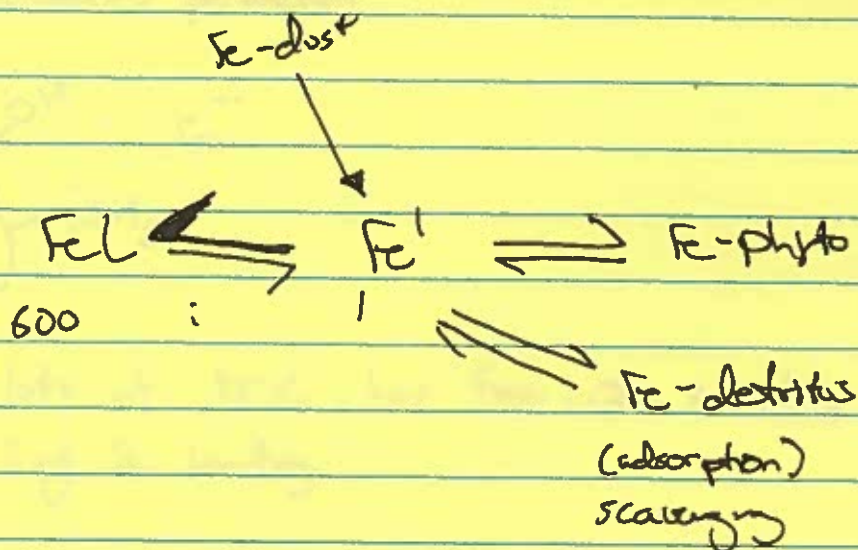
$$K \approx 10^{10} - 10^{13}$$

Forward rxn
might be strongly
favored

if $L = 2 \text{ nM}$ typical

$$\frac{\text{FeL}}{\text{Fe}^I} \approx 600$$

most of the iron in the oceans is bound to organic ligands



FeL expands the Fe inventory of the ocean

what is L?

unknown

2 forms of thought (East coast / West coast)

West
Coast

1) $\Rightarrow$ siderophores - organic molecules that microbes make in response to iron stress

- bind iron tightly
- help w/ uptake

but phytoplankton can't make them

↳ major Fe users

also [L] $\gg$ [siderophores]

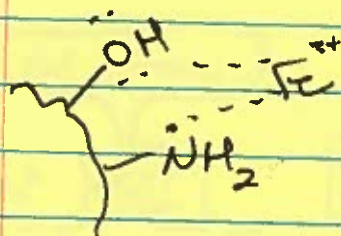
$\sim 2nM$

$\sim 10pM$

could be a methods problem

East
East

2) alot of organic molecules have affinity for iron
non-specific production



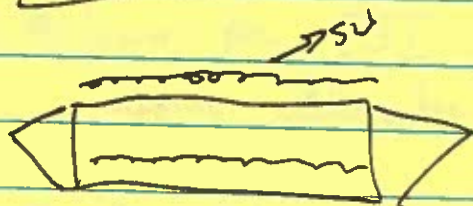
chemical
function
unit

lots of DOC has free -OH & -NH_2 & other moieties
w/ high Fe binding

-saccharides

-proteins

Iron limitation

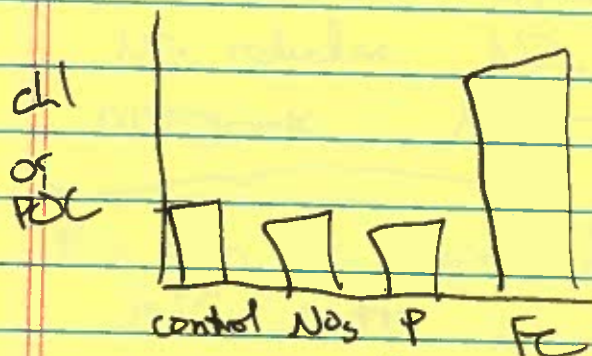


bottle assay

↓
SIS into abottle

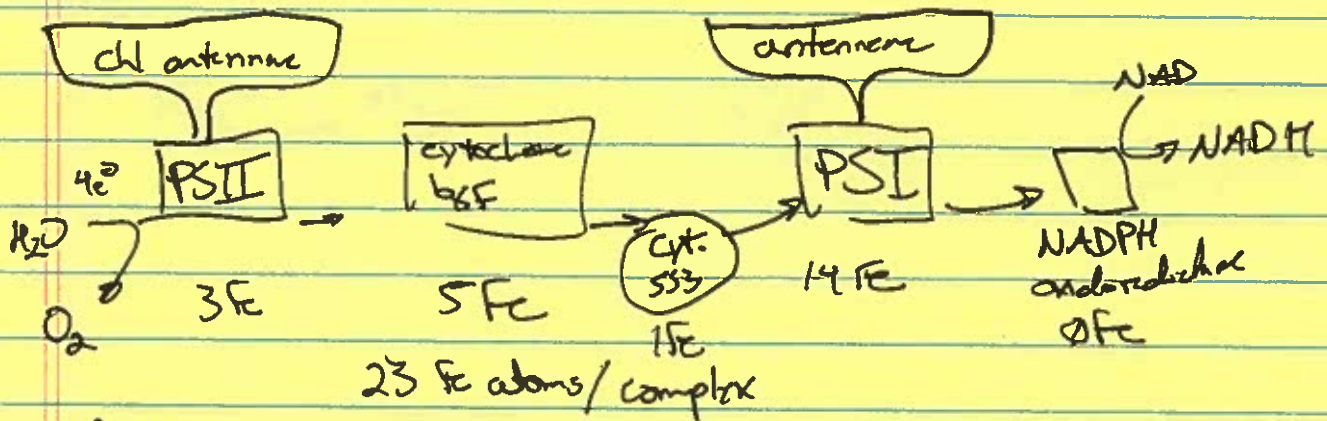
↓
add nutrient combinations

↓
harvest



if in Antarche

most of Fe use in cells is in photosynthesis



adaptations to Fe limitation

1) swap out unnecessary parts
 cyt $ss3$ (electron carrier) $\xrightarrow{\text{swapped}}$ plastocyanin
 Cu protein

2) & lower the PSI: PSII ratio
 require alternative electron flow away from PSI

other Fe-dependent enzymes

NO_3^- reductase $NO_3^- \rightarrow NH_4^+$

nitrogenase $N_2 \rightarrow NH_4^+$

This can be boiled down into an Fe quota
 Fe:C₁ ratio

$\rightarrow 1-10 \times 10^{-6}$ moles Fe: mol C

eg at Station Aloha if $1 \mu M$ C fixed/day ^{primary production}
 $1 \mu M \cdot (1-10 \times 10^{-6}) = 1 pM - 10 pM$ Fe/day